Notes: Chemmanur & Fulghieri (RFS, 1994)
Reputation, Renegotiation, and the Choice between Bank Loans and Publicly Traded Debt

Contents

1 Big picture

- **Question.** Why do some firms borrow from banks while other firms issue publicly traded debt?
- **Core idea.** Banks are repeated players in debt markets, so they can build a reputation for making the right renegotiation-versus-liquidation decision when borrowers enter financial distress. Bondholders are essentially one-period and dispersed, so they have weaker incentives to acquire costly information before deciding whether to renegotiate.
- **Main mechanism.** When a borrower is financially distressed, lenders can spend resources to learn whether the firm should be continued or liquidated. Banks care about future reputation and therefore invest more in evaluation. This lowers inefficient liquidation of viable firms.
- **Key trade-off.** Bank debt is more flexible and reduces inefficient liquidation, but it requires paying a higher interest rate to compensate the bank for reputation-building and information-production incentives.
- **Debt choice result.** Firms with a high probability of financial distress optimally choose bank loans; firms with a low probability of distress choose publicly traded debt.
- **Reputation result.** A bank with a stronger reputation devotes more resources to evaluation and charges a higher equilibrium loan rate.
- **Efficiency result.** Bank debt improves ex post efficiency relative to public bonds because banks are less likely to liquidate good but temporarily distressed firms.
- **Economic takeaway.** The value of bank debt is not only screening before lending or monitoring after lending. It is also credible, reputationally motivated renegotiation during distress.

2 Incremental contribution of the paper

- **From information advantage to reputation.** Much earlier banking theory emphasizes that banks have private information or superior monitoring technology. Chemmanur and Fulghieri deliberately do not assume that banks have private information unavailable to bondholders. Instead, banks' advantage comes from repeated-market reputation.
- **Renegotiation as the key difference.** The paper shifts attention from ex ante monitoring to ex post renegotiation. The bank's role is valuable when a borrower is distressed and lenders must decide whether to liquidate or renegotiate.
- **Endogenous information production.** The amount of resources lenders spend evaluating distressed firms is endogenous. Banks invest more because good renegotiation decisions build reputation and future rents.
- **A theory of debt-source choice.** The paper explains why high-distress-probability firms use bank loans and low-distress-probability firms use public debt. This prediction differs from pure asymmetric-information stories.
- **A theory of bank reputation.** Bank reputation is not a vague brand name. It is the market's belief that the bank is a low-cost evaluator and therefore likely to make correct continuation/liquidation decisions.
- **Joint explanation of rates and flexibility.** The model explains why bank loans may carry higher rates and still be optimal: the higher rate buys more efficient renegotiation and lower liquidation risk.
- **Empirical implications.** The paper generates testable predictions about borrower risk, bank reputation, renegotiation outcomes, interest rates, and the relative use of bank debt versus public debt.

Interpretation: The incremental contribution is to model banks as long-lived reputational intermediaries whose comparative advantage lies in credible renegotiation decisions, not simply in having better ex ante information.

3 Model primitives and measurement

This is a theoretical paper. The relevant "data and measurement" objects are the model primitives: project type, distress probability, liquidation value, lender information production, bank reputation, and interest rates.

3.1 Borrowers and project payoffs

Each entrepreneur has one project requiring one dollar of financing. The project succeeds with cash flow x with probability p and is financially distressed with probability $1 - p$. There are two borrower/project types:

$$p_U < p_S,$$

where type U is riskier and type S is safer. A fraction ϕ of borrowers are type S.

If a firm is financially distressed, it may still be a good firm. A fraction δ of distressed firms are good and can produce kx if allowed to continue under renegotiation, where $k \in (0, 1)$. The remaining fraction $1 - \delta$ are bad and produce zero if allowed to continue. Any firm can be liquidated for value y.

Interpretation: Financial distress does not necessarily mean economic failure. The model is about distinguishing good distressed firms from bad distressed firms.

3.2 Lenders and evaluation

Lenders can spend resources to evaluate a distressed borrower. The evaluation produces a signal that helps decide whether to liquidate or renegotiate. Let $q(c)$ denote the probability or informativeness of a good signal when the lender spends resources c, with higher c improving evaluation quality.

$$q'(c) > 0, \qquad q''(c) < 0.$$

Interpretation: The evaluation technology is costly and has diminishing returns. The lender must choose how much to invest in learning whether the distressed firm is worth saving.

3.3 Public bondholders

Public bondholders are one-period players. They do not have a continuing reputation to protect. They choose an interest rate and an evaluation level to maximize current expected payoff.

$$\pi_D^T(c) = y + \delta(kx - y)q(c) - c.$$

Interpretation: Bondholders may still evaluate, but their incentive is limited to the current debt claim. They do not obtain future reputation benefits from making the right renegotiation decision.

3.4 Banks

Banks are multiperiod players. A bank can be low-cost or high-cost in evaluation. Entrepreneurs do not observe the bank's type directly. Instead, they hold a belief:

$$\alpha_t = \Pr(\text{bank is low-cost at time } t).$$

This belief is the bank's reputation. A bank that makes better renegotiation/liquidation decisions in period 0 improves its reputation and can charge more favorable terms in period 1.

Interpretation: Bank reputation is endogenous. It is built by costly actions that borrowers cannot observe directly but whose consequences are eventually revealed.

3.5 Debt-source choice

The entrepreneur chooses between:

$$\text{bank loan} \quad \text{or} \quad \text{publicly traded debt.}$$

This choice is made after observing the bank's loan rate and the public debt rate. The entrepreneur compares the higher cost of bank debt with the expected benefit of better ex post renegotiation.

Interpretation: The choice is a trade-off between financing cost and financial flexibility. The model predicts that this trade-off depends on distress probability.

4 Models and theoretical specifications

4.1 Sequence of events

The model has two dates, $t = 0$ and $t = 1$. Each date has the same basic sequence:

1. The bank chooses a loan interest rate.
2. The public debt rate is competitively determined after observing the bank's rate.
3. The entrepreneur chooses bank debt or publicly traded debt.
4. If the firm is not in distress, cash flows are realized and lenders are repaid.
5. If the firm is distressed, lenders choose how much to evaluate the firm and then either renegotiate or liquidate.
6. Final cash flows are realized; after $t = 0$ the true types of borrowers become known and bank reputation is updated.

Interpretation: The ordering matters. The bank's rate is observable and can signal its reputation; public debt is competitively priced; the entrepreneur chooses the debt source; then renegotiation decisions determine ex post efficiency.

4.2 Public debt contract

For public bondholders, the expected payoff from a distressed firm after spending c^T on evaluation can be written as:

$$\pi_D^T = y + \delta(kx - y)q^T(c^T) - c^T.$$

The optimal evaluation choice solves:

$$\delta(kx - y)q'(c^{T*}) = 1,$$

when an interior optimum exists.

Interpretation: Public bondholders invest in evaluation only when the current expected benefit exceeds the current cost. They do not internalize future reputation benefits.

4.3 Bank loan contract

For a high-cost bank, the evaluation choice includes reputational benefits. A simplified way to express the first-order condition is:

current marginal benefit of evaluation + future reputational marginal benefit = marginal cost.

$$\delta(kx - y)q'(c^{B*}) + \Delta V'_{\text{reputation}}(c^{B*}) = 1.$$

Interpretation: The reputational term is the bank's advantage. Because a correct renegotiation decision improves future loan-market opportunities, a bank spends more on evaluation than dispersed bondholders.

4.4 Equilibrium with reputation acquisition

In equilibrium, entrepreneurs form beliefs about bank type. High-cost banks may imitate low-cost banks by choosing costly evaluation effort because future reputation is valuable. Low-cost banks naturally make better evaluation decisions. The market uses observed outcomes to update α_t.

$$\alpha_1 = \Pr(\text{low-cost bank} \mid \text{period-0 outcomes}).$$

Interpretation: Even high-cost banks have incentives to behave well if future reputation rents are large enough. This is the reputational discipline mechanism.

4.5 Robustness to out-of-equilibrium signalling

The paper considers whether low-cost banks can credibly reveal their type through out-of-equilibrium interest rates or renegotiation actions.

$$\text{No credible separating move} \quad \Rightarrow \quad \text{pooling or reputational equilibrium survives.}$$

Interpretation: This matters because if low-cost banks could simply reveal their type costlessly, reputation would be less important. The paper shows that such deviations are not credible under the model's payoff structure.

4.6 Bank loans versus public debt

The main comparison is between equilibrium bank debt and public debt.

- Bank loan yields are higher than public debt yields.
- Banks devote more resources to evaluating distressed firms than bondholders.
- Bank debt improves ex post efficiency by reducing inefficient liquidation of good distressed firms.
- The benefit of bank debt is larger for firms with higher probability of distress.

Interpretation: Bank loans are more expensive but more flexible. Public debt is cheaper but less effective in distress.

4.7 Borrower self-selection

Let U denote risky borrowers and S safe borrowers. Since $p_U < p_S$, type U firms are more likely to face distress and therefore value bank renegotiation more. Type S firms are less likely to need renegotiation and prefer cheaper public debt.

$$\Pr(\text{choose bank loan} \mid U) > \Pr(\text{choose bank loan} \mid S).$$

Interpretation: The model predicts that bank borrowers should look riskier than public debt issuers, not because banks caused risk, but because risky borrowers benefit more from bank flexibility.

4.8 Bank reputation and loan rates

If a bank's reputation α_0 increases, borrowers expect better renegotiation decisions. This has two effects:

- the bank invests more in evaluation when reputation rents are valuable;
- borrowers are willing to pay higher loan rates to more reputable banks.

Interpretation: A high-reputation bank can charge a higher rate because borrowers value the option to be efficiently renegotiated in distress.

5 Identification and theoretical design

This is not an empirical causal-design paper. Its "identification" is theoretical: the paper isolates the mechanism that distinguishes bank loans from public debt.

5.1 What the model holds fixed

The paper deliberately holds fixed or abstracts from several standard banking advantages:

- banks do not have ex ante private information that bondholders cannot access;
- bondholders are not assumed to have prohibitively high monitoring costs;
- the firm's project and distress structure are common across debt sources;
- the key difference is the horizon of the lender and its reputation motive.

Interpretation: This design isolates reputation for renegotiation as the marginal source of bank value. That is why the paper's contribution is distinct from monitoring and adverse-selection theories of bank debt.

5.2 Core theoretical comparison

The paper compares two financing technologies:

$$\text{public debt: one-period lenders with no reputation}$$

versus

$$\text{bank loans: repeated lender with reputation for correct renegotiation.}$$

Interpretation: The comparison asks what happens when a firm needs ex post flexibility. Public debt is cheaper ex ante but less flexible ex post; bank debt is more expensive but provides more credible renegotiation.

5.3 Why the repeated-game structure matters

If banks made only one loan, they would behave like public bondholders. The bank's advantage arises because the period-0 treatment of distressed borrowers affects period-1 reputation and loan rates.

$$\text{evaluation in period 0} \Rightarrow \text{correct renegotiation/liquidation} \Rightarrow \text{higher reputation } \alpha_1 \Rightarrow \text{future loan rents.}$$

Interpretation: Repetition transforms an otherwise unobservable evaluation decision into an incentivized reputational investment.

5.4 Theoretical falsifiable predictions

The model generates empirical predictions:

1. firms with higher expected distress probability should prefer bank loans;
2. public debt users should be safer on average;
3. more reputable banks should charge higher loan rates;
4. bank loans should be associated with fewer inefficient liquidations in distress;
5. bank evaluation effort should be greater than bondholder evaluation effort;
6. bank debt should be especially valuable when liquidation value is low relative to continuation value.

Interpretation: These predictions are not estimated in the paper, but they define the paper's empirical content and its connection to the bank-debt literature.

5.5 Main limitations

- The model abstracts from bank monitoring before distress.
- Bargaining over renegotiation outcomes is simplified.
- Bondholders are modeled as one-period players and cannot easily coordinate reputation.
- Bank choice is not modeled across competing banks; the firm chooses between one bank and public debt.
- The model is theoretical and does not quantify how large the reputation effect is in data.

Interpretation: The simplifications are deliberate. They isolate the mechanism of reputation-driven renegotiation.

6 Tables and figures

6.1 Figure 1: Uncertainty structure and project payoff; Table 1: The sequence of events

Read:

- Figure 1 shows the project tree. A firm succeeds with probability p and pays x. With probability $1-p$ it enters distress.
- In distress, a fraction δ of firms are good and should be renegotiated; a fraction $1-\delta$ are bad and should be liquidated.

- Table 1 lays out the two-period timing: bank rate, public debt rate, entrepreneur debt choice, cash-flow realization, evaluation and renegotiation/liquidation, and reputation updating.

Economic message: The figure defines the ex post decision problem. The table explains why banks have reputation: the period-0 decision affects period-1 beliefs and rents.

Detailed interpretation: The important reading point is the distinction between financial distress and economic inefficiency. Some distressed firms should continue. Lender evaluation determines whether the lender identifies those firms and renegotiates rather than liquidating them.

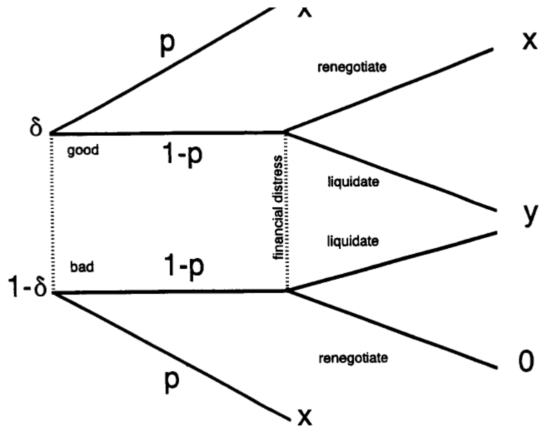

Figure 1
Uncertainty structure and project payoff
Each firm (project) yields a cash flow of x with probability p, $p \in \{p_l, p_h\}$; the firm will be in financial distress with the probability $(1-p)$. The probability p of a given firm is information private to the entrepreneur (firm insiders). Of the firms in financial distress, a fraction δ are "good" (i.e., they

(a) Figure 1: Uncertainty structure and project payoff.

The sequence of events

At time 0
1. Banks choose the interest rate at which to lend to entrepreneurs.
2. After the interest rate chosen by the bank is publicly observed, the interest rate in the bond market is competitively determined.
3. Entrepreneurs decide whether to borrow from the bank or to issue publicly traded debt.
4. Project cash flows are realized if the firm is not in financial distress, and the lenders are repaid in full with interest.
5. If the firm is in financial distress, lenders decide on the amount of resources to be devoted to evaluate the firm. Based on the evaluation, the firm is either liquidated or allowed to continue under a renegotiated debt contract.
6. Final project cash flows are realized, and the true types of all borrowing firms become known.

At time 1
1. Banks choose the interest rate at which to lend to entrepreneurs.
2. After the interest rate chosen by the bank is publicly observed, the interest rate in the bond market is competitively determined.
3. Entrepreneurs decide whether to borrow from the bank or to issue publicly traded debt.
4. Project cash flows are realized if the firm is not in financial distress, and the lenders are repaid in full with interest.
5. If the firm is in financial distress, lenders decide on the amount of resources to be devoted to evaluate the firm. Based on the evaluation, the firm is either liquidated or allowed to continue under a renegotiated debt contract.
6. Final project cash flows are realized.

(b) Table 1: The sequence of events.

Figure 1 and Table 1. The figure shows the economic problem and the table shows the timing. Together they explain why renegotiation reputation matters: distressed projects require information, and banks care about future beliefs after the first-period decision.

6.2 Table 2: Numerical illustration of the model for square root evaluation cost function $q(c) = 6\sqrt{c}$

Read:

- Table 2 presents three numerical cases with different probabilities of success, liquidation values, and payoff parameters.
- For each case, the table reports equilibrium interest rates, evaluation levels, reputation levels, and conditions needed for equilibrium existence.
- The table compares two initial bank reputation values, $\alpha_0 = 0.1$ and $\alpha_0 = 0.2$.
- Across cases, the bank's evaluation level exceeds the public bondholders' evaluation level, illustrating the reputation mechanism.

Economic message: The numerical table makes the theory concrete. It shows that the conditions for equilibrium can be satisfied under plausible parameter values and that banks devote more resources to evaluation than bondholders.

Detailed interpretation: The table is especially useful because the model is otherwise abstract. It shows how higher bank reputation changes loan rates and evaluation incentives, and how the relative benefit of bank debt varies with borrower risk.

Numerical illustration of the model for square root evaluation cost function ($q(c) = 6\sqrt{c}$

	Case 1		Case 2		Case 3	
Parameter values	$p_U = .8$ $p_S = .9$ $x = 7$ $y = .99$ $\delta = .1$ $k = .15$ $\bar{q} = .9$ $\phi = .4$		$p_U = .4$ $p_S = .6$ $x = 2.5$ $y = .98$ $\delta = .35$ $k = .4$ $\bar{q} = .6$ $\phi = .4$		$p_U = .3$ $p_S = .4$ $x = 5.5$ $y = .98$ $\delta = .25$ $k = .2$ $\bar{q} = .7$ $\phi = .3$	
α_0	.1	.2	.1	.2	.1	.2
$\underline{\alpha}_1$	.012	.012	.055	.055	.029	.029
$\underline{\alpha}_0$	0	0	0	0	0	0
$\bar{\alpha}_0$	.546	.559	.454	.469	.515	.514
$Q(\alpha_0)$	+	+	+	+	+	+
$q_t^{T*} = q_1^{H*}$	.108	.108	.126	.126	.540	.540
R_t^{T*}	1.002	1.002	1.021	1.021	1.024	1.024
R_1^{B*} (r)	1.046	1.065	1.098	1.144	1.074	1.119
R_1^{B*} (g)	1.004	1.006	1.045	1.073	1.055	1.088
R_1^{B*} (b)	1.014	1.027	1.064	1.103	1.067	1.109
R_1^{B*} (n)	1.014	1.025	1.059	1.096	1.065	1.106
α_1^r	.372	.537	.205	.330	.121	.231
α_1^g	.014	.037	.065	.140	.074	.156
α_1^b	.107	.021	.113	.220	.104	.206
q_0^{H*}	.169	.194	.258	.305	.566	.581
R_0^{B*}	1.022	1.036	1.152	1.208	1.125	1.191

p_U and p_S are the success probabilities of type U and type S firms, respectively. ϕ is the proportion of type S firms. x is the cash flow from each firm if it succeeds, and y if it is liquidated. δ is the prior probability that a firm in financial distress is good. $q_t(c)$ is the probability that a lender devoting an amount of resources c at time t to evaluate a firm in financial distress receives a good signal, conditional on the firm being truly good ($\bar{q}$ is the highest achievable value of this probability; q_t^{T*} and q_t^{H*}, $t \in \{0, 1\}$, denote the equilibrium values of this probability for bond holders and for the high-cost bank, respectively). k is the fraction of the payoff from a firm in financial distress (allowed to continue under debt renegotiation) retained by lenders. R_t^{T*} and $R_t^{B*}(s)$ are the equilibrium interest rates charged by bond holders and by the bank, respectively, at time t, $t \in \{0, 1\}$, $s \in \{r, g, b, n\}$. α_0 and α_1^s are the reputation levels of the bank at time 0 and time 1, respectively. The conditions $Q(\alpha_0) \geq 0$, $\alpha_0 \geq \underline{\alpha}_0$ and $\alpha_1^s \geq \underline{\alpha}_1$ need to be met for the existence of equilibrium.

6.3 Model-result box: Reputation and evaluation effort

Mechanism	Private incentive	Economic implication
Public debt	Bondholders maximize current payoff from evaluation.	Evaluation is limited to current-claim benefits.
Bank debt	Banks internalize future reputation rents.	Banks spend more on evaluation and reduce inefficient liquidation.
Higher reputation	Borrowers expect better renegotiation decisions.	Reputable banks can charge higher loan rates.
High distress risk	Borrowers expect to need renegotiation more often.	Riskier firms choose bank loans.
Low distress risk	Borrowers rarely need renegotiation.	Safer firms choose cheaper public debt.

Notes-created result table: Reputation and evaluation effort. This table summarizes the main theoretical mechanism; it is not an original table from the paper.

7 Short summary

- The paper explains the choice between bank loans and public debt.
- Banks are repeated players and can build reputation for correct renegotiation.
- Public bondholders are one-period lenders and lack the same reputation incentives.
- When borrowers enter financial distress, lenders must decide whether to liquidate or renegotiate.
- Some distressed firms are good and should be continued.
- Banks invest more in costly evaluation because their current decisions affect future reputation.
- Bank debt is more expensive but more flexible.
- Public debt is cheaper but involves higher risk of inefficient liquidation.
- High-distress-probability firms prefer bank loans.
- Low-distress-probability firms prefer publicly traded debt.

- Reputable banks can charge higher loan rates.
- Bank loans improve ex post efficiency by reducing inefficient liquidation.

8 Conclusion

8.1 Core conclusion

Chemmanur and Fulghieri show that bank loans can dominate public debt when borrowers may need efficient renegotiation in financial distress. Banks are valuable because they are long-lived and can build a reputation for making correct liquidation-versus-renegotiation decisions.

8.2 Economic intuition

Public bondholders face a one-shot decision. They may spend too little on evaluating a distressed firm because they only receive current-period benefits from avoiding inefficient liquidation. Banks face a different incentive. If a bank evaluates carefully and makes the right decision, future borrowers update beliefs about the bank and are willing to pay for its renegotiation services. This future reputation rent makes the bank spend more today.

8.3 Incremental contribution revisited

The paper's most important contribution is to offer a theory of bank debt that does not rely on banks having superior private information at origination. Instead, banks' comparative advantage is dynamic: reputation makes them more credible and disciplined renegotiators. This shifts the bank-versus-bond comparison from ex ante monitoring to ex post financial flexibility.

8.4 Implications for corporate finance

The model explains why firms with different distress probabilities choose different debt sources. Riskier firms value bank flexibility and choose bank loans despite higher rates. Safer firms do not expect to need renegotiation and choose cheaper public debt. This provides a theoretical foundation for empirical patterns in which bank borrowers are more opaque, riskier, or more likely to need financial flexibility.

8.5 Implications for bank reputation

Reputation is a productive asset. A reputable bank can charge a higher rate because borrowers expect it to devote more resources to evaluation in distress. The paper therefore predicts that reputation affects both the quantity and price of bank lending.

8.6 Limitations and open questions

The model abstracts from many features of bank lending, including ex ante screening, monitoring covenants, relationship lending, renegotiation bargaining, multiple competing banks, and the possibility that bondholders coordinate through trustees. It also does not empirically measure reputation or renegotiation. These omissions make the model clean but leave room for empirical tests and richer institutional models.

8.7 Takeaway

The paper's takeaway is that the value of bank debt is tied to what happens when things go wrong. Banks do not only finance projects; they preserve value in distress by using reputation to commit to careful renegotiation and liquidation decisions.